Privacy-Enhanced Synthetic Data Generation for Autonomous Vehicle Training

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Project Proposal Report

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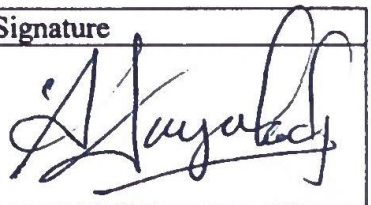
March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Autonomous vehicle (AV) strongly depends on lot of sensor information to practice machine learning models used for perception, object findings, and decision making, normal datasets used in AV project often have sensitive data like, pedestrian faces, vehicle license plates, and location captured from real world natures. The availability of such data gives significant safety risks and legal threats, specifically under international data security like the GENERAL DATA PROTECTION REGULATION (GDPR). In the last few years, simulation platforms like AirSim have been mostly used to create synthetic datasets that lower the need for large space real world data collection. While synthetic data generation assists address security concerns, there still uncertainty regarding whether practiced techniques may still show information about the datasets used while the training process.

A major project gap in remaining synthetic data pipeline is lack of systematic privacy assurance methods capable of detecting whether security saving techniques effectively stop information exploitation. Many remaining startups focus on starting security methods but give limited empirical validation against real world attack threats. This project component issues this limitation by improving a security assurance module that detects potential security leakage in machine learning models practiced using synthetic autonomous vehicle datasets.

The proposed startups focuses on starting membership inference attacks to detect whether an attacker can catch if a particular data record was included in the model's practicing data set. The attack framework will be proceeded using python-based machine learning tools like PyTorch and scikit learning, managing shadow models and attack classifiers to simulate real world adversarial scenarios. Security risk will be assured using metrics like attack precision, accuracy, recall and inference success rate.

The results of this project will give empirical proof regarding the effectiveness of security saving models used inside the synthetic information generation pipeline by validating the robustness of practiced models against inference threat , this component contributes to the overall research by giving a privacy verification layer that assists the improvement of security aware and regulatory compliant AI program for autonomous vehicle applications.

Keywords: Membership Inference Attack, Autonomous Vehicle Data, Synthetic Data Privacy, Machine Learning Security, GDPR

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TABLE OF CONTENT

1.INTRODUCTION 8
1.1 Background and Context 9
1.2 Literature Review 10
1.3 Research Gap 12
2.OBJECTIVE 14
2.1 Main Objective 15
2.2 Specific Objectives 15
3. METHODOLOGY 15
4.HIGH-LEVEL SYSTEM ARCHITECTURE 16
5.USER REQUIRENMENTS 18
Stakeholders 19
Functional Requirements 19
Non-Functional Requirements 19
6.COMMERCIALIZATION PLAN 20
6.1 Target Market and Customer Persona 21
6.2 Value Proposition 21
6.3 Cost Estimation 21
6.4 Competitive Advantages 21
7. BUDGET AND JUSTIFICATION 21
Budget 22
Justification 22
8. WORK BREAKDOWN STRUCTURE 23
8.1 Timeline Alignment 24
09. GANTT CHART 25
10. REFERENCES 26
11. APPENDICES 27
Appendix A – Risk Analysis 28
Appendix B – System Architecture Overview 28
Appendix C – Experimental Environment 28

LIST OF TABLES

Table 1: The Budget.....20

Table 2: Task Breakdown per Member.....22

Table 3: Timeline Alignment.....23

LIST OF FIGURES

Figure 1: Data Flow Diagram.....15
Figure 2: High level System Architecture.....16

LIST OF ABBREVIATIONS

AV	Autonomous Vehicle
DP	Differential Privacy
DP-SGD	Differentially Private Stochastic Gradient Descent
GAN	Generative Adversarial Network
GDPR	General Data Protection Regulation
GPU	Graphics Processing Unit
IoU	Intersection over Union
LiDAR	Light Detection and Ranging
mAP	Mean Average Precision
MIA	Membership Inference Attack
VAE	Variational Autoencoder
YOLO	You Only Look Once (object detection model)
SDG	Sustainable Development Goal
AIMS	Autonomous Intelligent Machines and Systems
PDPA	Personal Data Protection Act
NIST	National Institute of Standards and Technology
LKR	Sri Lankan Rupee

1.INTRODUCTION

1.1 Background and Context

Autonomous vehicles represent one of the most significant technological developments in modern transportation systems. These systems rely heavily on machine learning models that analyze sensor data such as images, LiDAR readings, and radar signals to understand the surrounding environment and make driving decisions.

Training such models requires large volumes of high quality data collected from real world driving environments. Popular datasets such as KITTI, Waymo Open Dataset, and nuScenes provide annotated sensor data that researchers used to train perception and object detection models. However, these datasets frequently capture sensitive personal information including pedestrian faces, vehicle license plates, and detailed location paths.

The presence of such personal data introduces significant privacy concerns. Regulations such as the General Data Protection Regulation (GDPR) in the European Union impose strict requirements on how personal data can be collected, processed, and stored. Organizations developing autonomous vehicle technologies must therefore ensure that their data processing pipelines comply with privacy and security standards.

The creation of synthetic data has become a promising alternative for addressing these challenges. Simulation environments such as CARLA, AIRSim allow developers to generate realistic driving scenarios while avoiding direct collection of personal data. These platforms enable the creation of diverse traffic conditions, weather scenarios, and edge cases that are difficult to capture through real-world data collection.

Despite these advantages, synthetic data pipelines may still introduce privacy risks. Models for machine learning that were trained on datasets that incorporate real world calibration data may unintentionally memorize sensitive patterns present in the original data. If an attacker can exploit such patterns, they may infer whether records were used during model training.

One of the most significant privacy threats in this context is the Membership Inference Attack (MIA). This attack attempts to determine whether a specific data sample was included in the training dataset of a machine learning model. If successful, such attacks can reveal sensitive information about individuals present in the original dataset.

This research focuses on evaluating such privacy risks through security testing techniques. By implementing attack based evaluation methods, this component aims to verify whether privacy preserving mechanisms effectively protect training data within the synthetic data generation framework.

The research also aligns with several United Nations Sustainable Development Goals (SDGs), particularly SDG 9 (Industry, Innovation, and Infrastructure) and SDG 11 (Sustainable Cities and Communities), by promoting safe and responsible deployment of intelligent transportation technologies.

1.2 Literature Review

The development of privacy preserving synthetic datasets for autonomous vehicle (AV) systems has been influenced by several areas of research, including differential privacy, generative models, simulation platforms, and machine learning security. Previous studies provide important theoretical and technical foundations that support the approach proposed in this research.

One of the most important concepts in privacy preserving data processing Differential Dwork et al. introduced privacy.. [7]. This framework provides a mathematical guarantee that an algorithm's output will not significantly change whether or not The dataset contains information on a certain person. In practice, This is accomplished by incorporating precisely regulated noise into the learning process. Mechanisms such as the Gaussian and Laplace distributions are commonly used to inject this noise while maintaining a balance between data privacy and model accuracy.

Within the framework of synthetic data generation, deep generatives models have played a key role. Goodfellow et al. [8] introduced Generative Adversarial Networks (GANs), which comprise two neural networks that compete with one another to produce realistic synthetic data. GANs have since been widely used for generating synthetic sensor data in machine learning applications. Another popular generative approach is the Variational Autoencoder (VAE) proposed by Kingma and Welling [9], which provides a structured latent representation that allows for controlled data generation. These generative techniques form the basis for many synthetic dataset generation pipelines used in autonomous vehicle research.

Building upon the concept of differential privacy, Abadi et al. [10] developed the algorithm known as Differentially Private Stochastic Gradient Descent (DP-SGD). This technique enables deep learning models to be trained while maintaining differential privacy guarantees. The algorithm works during training, by removing slopes and increasing Gaussian noise before updating model parameters. As a result, the influence of any individual data record on the trained model is limited. DP-SGD has become one of the most widely adopted techniques for implementing differential privacy in systems for deep learning.

Several software libraries have been created to support practical implementation of these techniques. The Opacus library [11], developed by Meta AI, provides a PyTorch-based framework for training neural networks with differential privacy. Similarly, TensorFlow Privacy [12] offers a set of tools for integrating privacy-preserving mechanisms into TensorFlow-based machine learning models. Both libraries include privacy accounting mechanisms that track the privacy budget during model training, ensuring that privacy guarantees remain within acceptable limits.

In autonomous vehicle research, simulation platforms are widely used to generate large-scale datasets. Dosovitskiy et al. [5] introduced the CARLA simulator, an open-source urban driving

simulator that enables researchers to create configurable traffic environments and collect sensor data such as camera images and LiDAR readings. CARLA supports automated scenario generation and ground-truth annotations, making it a valuable tool for developing and testing AV perception systems.

Despite these advancements, Models for machine learning can still be vulnerable to privacy attacks. Shokri et al. [6] demonstrated the feasibility of Membership Inference Attacks (MIA), where an attacker attempts to determine whether a particular data record was utilized in the model training. Their work showed that machine learning models may unintentionally reveal information from prediction outputs regarding their training datasets,, highlighting the importance of privacy-preserving training techniques.

Further research has investigated how to incorporate differential privacy into complex machine learning environments. For example, Geyer et al. [13] applied differential privacy techniques to federated learning systems used in medical imaging. Their work demonstrated that privacy constraints can be successfully applied in distributed and large-scale machine learning pipelines while maintaining acceptable performance levels.

Recent studies have also highlighted the risks associated with autonomous driving datasets. Chen et al. [14] showed that even anonymized driving data can sometimes be re-identified through behavioral patterns and trajectory analysis. This finding emphasizes the need for stronger privacy guarantees when using such datasets for machine learning applications.

Another approach to privacy-preserving synthetic data generation was proposed by Jordon et al. [15], who developed PATE-GAN, a generative model that integrates differential privacy through a teacher student training mechanism. This method enables the generation of synthetic datasets while preventing direct exposure of sensitive training data.

Overall, the existing literature demonstrates significant progress in machine learning methods that protect privacy and generate synthetic data.. However, many studies primarily focus on developing privacy mechanisms rather than evaluating their effectiveness against real world privacy attacks. This highlights the importance of incorporating security testing methods, such as membership inference attacks, to verify whether privacy preserving approaches truly protect sensitive data.

1.3 Research Gap

Although significant progress has been made in both synthetic data generation and privacy preserving machine learning, several important limitations still remain in the context of autonomous vehicle (AV) data pipelines. Existing research highlights a number of areas where current approaches are insufficient, particularly when it comes to integrating strong privacy guarantees with synthetic data generation frameworks.

One major limitation is the lack of formal privacy guarantees in many existing synthetic data platforms used for autonomous vehicle development. Simulation environments such as CARLA and AirSim are primarily designed to generate realistic driving scenarios and diverse environmental conditions. While these platforms are highly effective for producing large datasets, they generally focus on realism and scalability rather than privacy protection. As a result, there is often no mathematically verifiable assurance that sensitive attributes originating from real world calibration datasets cannot be indirectly encoded or reproduced in the generated outputs.

Another issue is that privacy protection is often treated as a secondary step rather than being integrated into the system design itself. In many existing solutions, privacy techniques such as anonymization or masking are applied only after the data generation process is completed. However, these post processing methods do not fully address the risk that machine learning models may memorize sensitive patterns during training. If such memorization occurs, sensitive information may still be exposed through model outputs even after the data has been anonymized.

A further challenge lies in the absence of well defined evaluation frameworks for analyzing the trade off between privacy and data utility in autonomous vehicle systems. While differential privacy provides theoretical guarantees for protecting individual data records, its practical impact on the performance of AV perception models is not always clearly measured. Without standardized evaluation of metrics, it becomes difficult for researchers to determine how privacy constraints affect the accuracy and reliability of machine learning models used in real world driving scenarios.

Additionally, many existing synthetic data systems lack systematic validation against known privacy attack methods. For example, membership inference attacks and re identification attacks are widely recognized threats in machine learning security research. However, many proposed privacy preserving frameworks do not include empirical testing against these attacks. As a result, claims regarding the privacy protection of synthetic datasets often remain theoretical rather than experimentally verified.

To address these limitations, this research focuses on integrating privacy preserving techniques directly into the synthetic data generation pipeline while also evaluating the effectiveness of these mechanisms through security testing methods. In particular, the study emphasizes the importance of incorporating differential privacy during the model training process and validating its effectiveness using membership inference attack experiments.

The novelty of this work does not lie in the invention of new privacy algorithms, but rather in the practical integration and evaluation of existing privacy techniques within an autonomous vehicle data generation framework. By combining privacy by design principles with empirical attack based validation, this research aims to contribute to the development of more trustworthy and privacy aware machine learning systems for autonomous vehicle applications.

2.OBJECTIVE

2.1 Main Objective

The main objective of this research component is to design and implement a Membership Inference Attack (MIA) evaluation framework that can measure potential privacy leakage in machine learning models trained on synthetic autonomous vehicle datasets. The system Focus to simulate realistic attack scenarios where an adversary attempts to determine whether a specific data sample was included in the model's training dataset. By evaluating the success rate of these attacks, the framework will provide a practical method for assessing the effectiveness of privacy preserving mechanisms such as differential privacy. The results generated by this component will help identify privacy risks in trained models and support the development of safer and more trustworthy data generation pipelines for autonomous vehicle research.

To develop a privacy aware synthetic data framework for autonomous vehicle training that ensures data protection through privacy preserving techniques and security validation mechanisms.

2.2 Specific Objectives

- To analyze existing privacy attack models and identify potential threats to machine learning systems trained using synthetic datasets.
- To design and implement a Membership Inference Attack framework for evaluating privacy leakage in machine learning models.
- To measure the effectiveness of privacy preserving techniques by comparing attack success rates under different training configurations.
- To evaluate the robustness of synthetic autonomous vehicle datasets against inference attacks.
- To provide empirical evidence that supports the development of privacy preserving data generation pipelines.

3. METHODOLOGY

This research adopts an experimental evaluation methodology. The main objective is to simulate privacy attacks against machine learning models trained using synthetic datasets generated by the group's data pipeline.

Initially, synthetic datasets produced using the CARLA simulator will be obtained from the simulation component of the group project. These datasets will then be used to train machine learning models for perception tasks related to autonomous driving.

Once the models are trained, Membership Inference Attack models will be implemented. These attacks attempt to determine whether a specific data sample was used during the model training process. The attack will be implemented using the shadow model technique, where multiple models are trained to mimic the target model's behavior.

The attack framework will analyze model prediction outputs and identify patterns that distinguish between training and non-training samples. Machine learning classifiers will then be used to predict membership status based on these patterns.

The implementation will be developed using Python, with machine learning frameworks such as PyTorch and Scikit learning used for model training and attack simulation.

To evaluate privacy risks, several performance metrics will be used, including:

- Attack accuracy
- Precision and recall
- Attack success rate
- Privacy leakage indicators

The results will be analyzed to determine whether privacy preserving techniques implemented within the overall framework effectively reduce the success rate of inference attacks.

Ethical considerations will be addressed by ensuring that only publicly available datasets and synthetic data are used in the experiments. The research will also follow privacy principles defined in regulatory frameworks such as GDPR and NIST privacy guidelines.

4.HIGH-LEVEL SYSTEM ARCHITECTURE

The overall system architecture of the project consists of several interconnected components responsible for synthetic data generation, privacy protection, model training, and security evaluation. The simulation component generates synthetic autonomous vehicle datasets using the AirSim simulator. These datasets are then processed by the privacy preserving module, which includes differential privacy mechanisms during the model training process.

Each component of this research focuses on the Security Evaluation Module, which analyzes trained models to detect potential privacy leakage. Data flows from the dataset generation module to the model training component, after which the trained models are evaluated using the membership inference attack framework. The results of this evaluation help determine whether privacy protection mechanisms are effective.

The architecture is designed to ensure scalability, allowing the framework to handle large datasets and multiple training configurations. Security considerations include restricting access to sensitive datasets and ensuring that privacy evaluation results are properly documented.

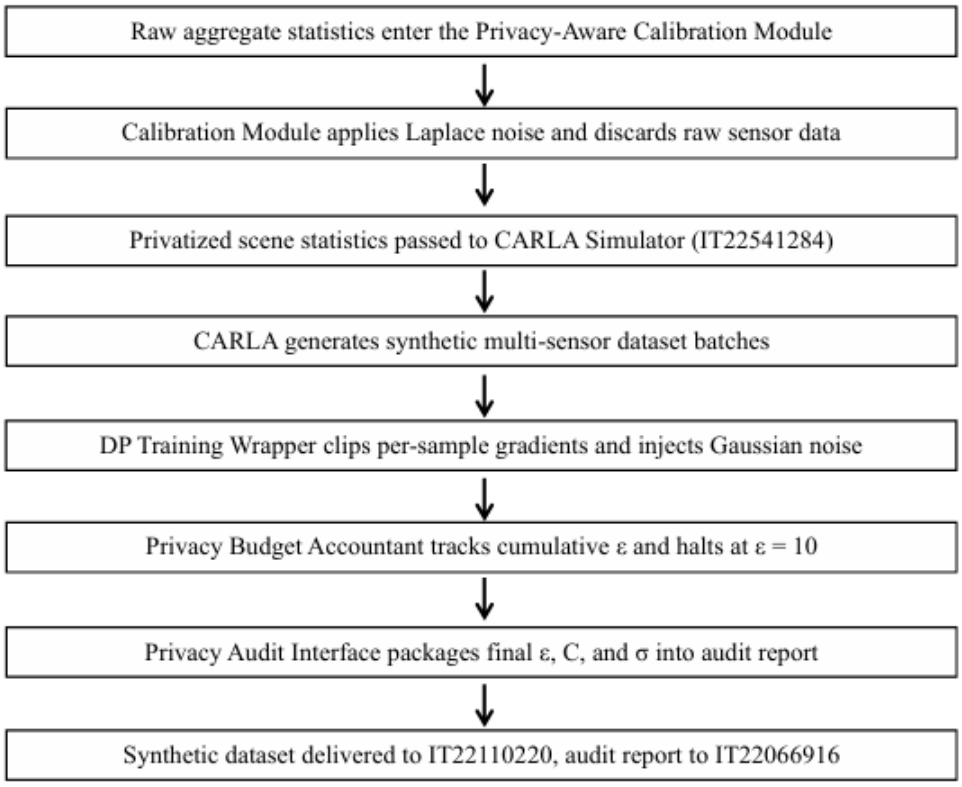


Figure 1 : Data Flow Diagram

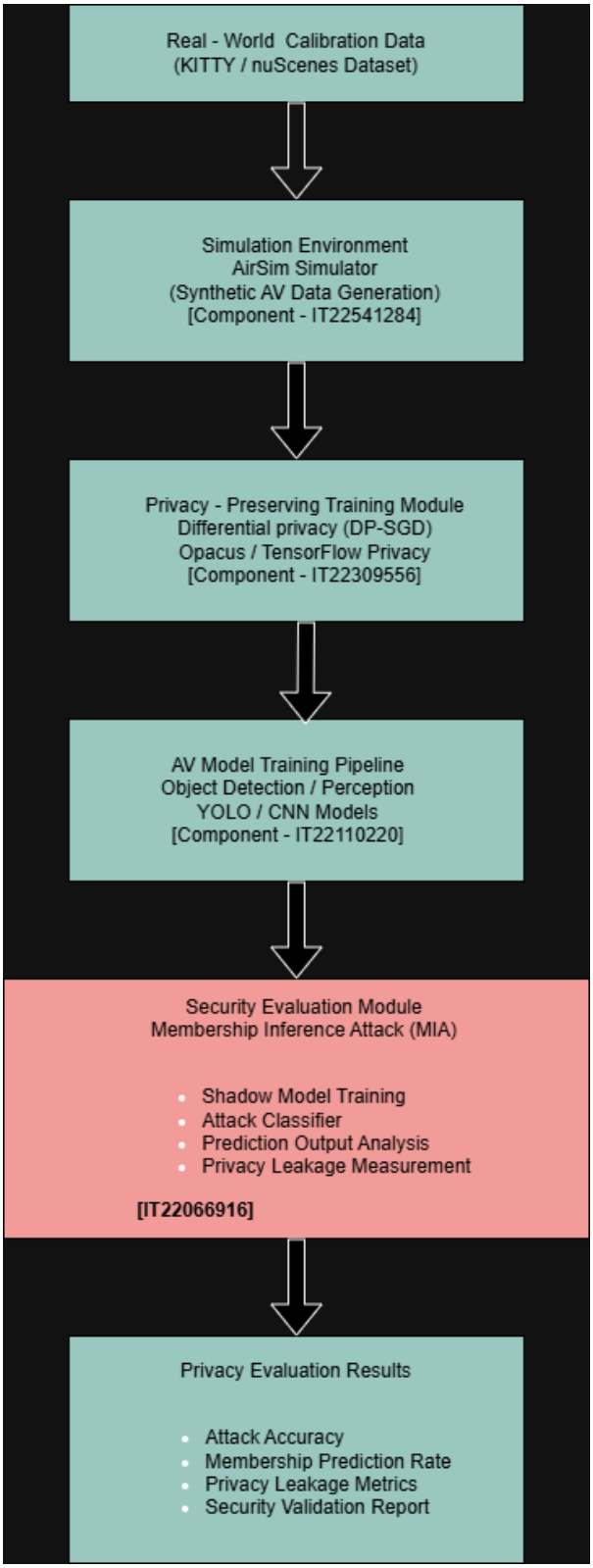


Figure 2 : High – Level System Architecture

5.USER REQUIRENMENTS

Stakeholders

Primary stakeholders interact directly with the security evaluation results produced by this component. Autonomous vehicle developers and machine learning engineers require reliable methods to verify whether trained models leak sensitive information from their training datasets. The Membership Inference Attack (MIA) testing module provides measurable evidence about potential privacy leakage in models trained using synthetic data. Within the research group, the outputs of this component help validate whether the privacy preserving techniques applied in earlier stages of the pipeline are effective. The academic supervisor is also a key stakeholder responsible for reviewing the research methodology and approving the evaluation results.

On the secondary side, data protection regulators and compliance officers may use such evaluation results to understand whether machine learning systems meet privacy requirements under regulations such as GDPR. Researchers working in AI security and privacy preserving machine learning may also benefit from this work by using the proposed attack evaluation framework to study privacy risks in synthetic data pipelines.

- Autonomous vehicle researchers
- Machine learning engineers
- Automotive industry developers
- Data protection officers
- Regulatory authorities

Functional Requirements

- **FR1** – The system shall simulate membership inference attacks on trained machine learning models.
- **FR2** – The system shall evaluate privacy leakage using attack performance metrics.
- **FR3** – The system shall analyze model outputs to detect potential information exposure.
- **FR4** – The system shall generate evaluation reports summarizing privacy risks.
- **FR5** – The system shall support testing under multiple privacy configurations.

Non-Functional Requirements

- **Security** – The system must ensure that sensitive datasets are protected during experimentation.
- **Performance** – Attack evaluation processes should operate efficiently on large datasets.
- **Usability** – The evaluation results should be easily interpretable for researchers and developers.
- **Scalability** – The framework should support testing across multiple machine learning models.

- **Compliance** – The system should align with GDPR, PDPA, and privacy guidelines.

6.COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

Organizations developing machine learning systems using sensitive datasets represent the primary target market for this component. Autonomous vehicle developers, AI research labs, and technology companies require reliable tools to verify whether their trained models leak information about the datasets used during training. Machine learning engineers and security analysts working in these organizations need practical methods to test models for privacy risks before deployment. A secondary market includes academic researchers and data protection compliance teams who need evaluation frameworks to study privacy risks and validate the effectiveness of privacy-preserving techniques.

6.2 Value Proposition

The proposed Membership Inference Attack (MIA) testing framework provides organizations with a practical method to evaluate whether their machine learning models unintentionally reveal information about the training dataset. By simulating realistic attack scenarios, the system produces measurable privacy risk indicators that help developers understand whether their models are vulnerable to inference attacks. This allows organizations to strengthen their privacy protection mechanisms and demonstrate compliance with data protection regulations.

6.3 Cost Estimation

The primary development costs for this component involve computing resources required for running machine learning experiments and attack simulations. Since the implementation relies mainly on open-source technologies such as Python, PyTorch, and Scikit-learn, software licensing costs remain minimal. Cloud computing and GPU processing represent the main operational expenses.

6.4 Competitive Advantages

Many existing synthetic data platforms focus mainly on generating datasets but do not include strong mechanisms for validating privacy risks. The proposed attack evaluation module provides a unique advantage by offering empirical privacy testing through membership inference attacks, allowing developers to verify the security of their machine learning models before deployment.

7. BUDGET AND JUSTIFICATION

Budget

Table 1 : The Budget

Item	Estimated Cost (LKR)	Justification
Personal Laptop	LKR 0	Development and testing run on an existing personal laptop. No new hardware purchase is required. Compute intensive training runs go to cloud GPU instances rather than a dedicated local GPU.
Cloud GPU Compute (AWS, approx. 200 hours)	LKR 35,000	DP training runs slower than standard training due to per-sample gradient computation. Multiple ablation experiments across different ε settings must be completed before the final generation run.
Cloud Storage (500 GB)	LKR 5,000	Storage for synthetic dataset batches, model checkpoints, and privacy budget logs across all experimental runs.
Opacus and TensorFlow Privacy (open source)	LKR 0	Both libraries are freely available under open source licenses and do not require any paid subscription or seat license.
Research paper access (IEEE Xplore, ACM DL	LKR 8,000	Key papers on differentially private GANs and AV data privacy sit behind paywalls and require institutional or individual purchase .
Human effort (research and implementation)	LKR 0	The academic project scope covers all research and implementation work. No external or commercial labor cost applies
Total	**LKR 48,000**	

Justification

The budget required for this component is relatively low because the implementation mainly depends on open-source software tools and publicly available datasets. Machine learning frameworks such as PyTorch, Scikit learn, and Opacus are freely available for academic research. The primary expense comes from computing resources needed to train models and perform membership inference attack simulations, which may require GPU processing. Additional minor costs include data storage and internet usage required for downloading datasets and running experiments. Overall, the budget mainly supports computational resources needed to conduct reliable experiments and evaluate privacy risks in machine learning models.

8. WORK BREAKDOWN STRUCTURE

Tasks are assigned to members according to their individual research focus. Phases, weeks, and key tasks for all four members are listed on the table.

Table 2 : Task Breakdown per Member

Member	Role	Phase	Weeks	Key Tasks
IT22309556	Privacy Integration	1	1-3	Literature review on DP mechanisms. Library selection (Opacus vs TF Privacy). Define ε range for ablation.
IT22309556	Privacy Integration	2	4-7	Build DP training wrapper. Implement Privacy Aware Calibration Module with Laplace noise. Run ablation experiments.
IT22309556	Privacy Integration	3	8-10	Full scale generation run under ε DP. Monitor ε consumption. Confirm budget not exceeded.
IT22309556	Privacy Integration	4	11-14	Coordinate with IT22066916 on MIA tests.
IT22541284	Dataset Generation / Simulation	1	1-3	Set up AirSim simulator environment. Define scene and sensor configurations.
IT22541284	Dataset Generation / Simulation	2	4-8	Run AirSim simulation using privatized calibration statistics from IT22309556. Generate raw synthetic dataset batches.
IT22541284	Dataset Generation / Simulation	3	9-12	Validate dataset format and sensor coverage. Deliver final synthetic dataset to IT22110220.
IT22110220	Utility Evaluation	1	1-4	Define utility metrics (mAP, FID). Set up AV model training pipeline and baseline benchmarks.
IT22110220	Utility Evaluation	2	8-11	Train AV model on synthetic dataset. Measure mAP and FID against non private baseline.
IT22110220	Utility Evaluation	3	12-14	Report utility trade-off results. Contribute utility analysis to group thesis.
IT22066916	Risk and Compliance	1	1-3	Review GDPR, CCPA, and Sri Lanka PDPA requirements. Define compliance checklist
IT22066916	Risk and Compliance	2	9-12	Run membership inference and re identification attack tests on DP synthetic data from IT22309556.
IT22066916	Risk and Compliance	3	12-14	Compile attack results and compliance report. Contribute risk and regulatory sections to group thesis.

8.1 Timeline Alignment

Shaded cells mark weeks where a member has active tasks. Phases overlap at inter-member handoff points: IT22309556 delivers privatized calibration statistics to IT22541284 at the end of Phase 1 and delivers the ε-DP audit log to IT22066916 at the start of Phase 3.

Table 3 : Timeline Alignment

Member	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
IT22309556	1	2	3	4	5	6	7	8	9	10	11	12	13	14
IT22541284	1	2	3	4	5	6	7	8	9	10	11	12		
IT22110220	1	2	3	4				8	9	10	11	12	13	14
IT22066916	1	2	3						9	10	11	12	13	14

09. GANTT CHART

Task	Semester 1 (Mar – May 2026)			Semester 2 (Jun – Dec 2026)						
All Members	**Mar**	**Apr**	**May**	**Jun**	**Jul**	**Aug**	**Sep**	**Oct**	**Nov**	**Dec**
Literature Deep Dive and Architecture Mapping										
Hardware and Storage Allocation Setup										
Calibration Strategy and Integration Checkpoint										
Full Pipeline Execution and QA Review										
Metric Compilation and Final Paper Drafting										
SLIIT ASSESSMENT MILESTONES										
★ Proposal Submission (15 Mar 2026)	★									
★ Proposal Presentation (16–18 Mar 2026)	★									
★ Progress Presentation 1 (4–6 May 2026)			★							
★ Progress Presentation 2 (17–19 Aug 2026)						★				
★ Final Report Submission (11 Oct 2026)								★		
★ Final Presentation and Viva (12–14 Oct 2026)								★		
★ Research Paper Submission (15 Oct 2026)								★		
★ Publication Evidence Submission (1 Dec 2026)										★

Figure 2 : GANTT CHART

Star cells mark SLIIT official milestones. Semester 1 runs from March to May 2026; Semester 2 runs from June to December 2026.

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11. APPENDICES

Appendix A – Risk Analysis

During the development of the proposed system, several risks may affect the research progress and outcomes. One possible risk is the limited availability of computational resources, since membership inference attack experiments require training multiple shadow models and performing repeated evaluations. Another risk is the difficulty in achieving accurate attack results, as the success rate of membership inference attacks depends on the model structure and training configuration. Additionally, privacy preserving mechanisms such as differential privacy may reduce model performance, which could affect the reliability of the evaluation results. To reduce these risks, the project will use optimized machine learning libraries, controlled experimental setups, and multiple evaluation trials to ensure consistent and reliable results.

Appendix B – System Architecture Overview

The proposed system architecture integrates multiple components developed by different group members to create a complete privacy-aware autonomous vehicle data pipeline. Synthetic datasets generated from the simulation environment are first processed through the privacy preserving training module, where differential privacy mechanisms are applied during model training. The trained models are then evaluated using the Membership Inference Attack testing module, which represents the focus of this research component. This module simulates adversarial attempts to determine whether a particular data record was included in the training dataset. The evaluation results are used to measure the privacy leakage risk of the trained models and verify the effectiveness of the applied privacy preserving techniques.

Appendix C – Experimental Environment

The experiments in this research will be conducted using commonly used machine learning frameworks and tools. The implementation will primarily rely on Python based libraries such as PyTorch, Scikit learn, and Opacus for model training and differential privacy integration. Synthetic datasets generated from the AirSim autonomous driving simulator and publicly available datasets such as the Waymo Open Dataset will be used for training and evaluation. The experiments will be executed on a workstation environment with GPU acceleration to support model training and attack simulation. The performance of the attack framework will be evaluated using metrics such as attack accuracy, precision, recall, and privacy risk indicators.